



Indian Association for the Cultivation of Science
(Deemed to be University under *de novo* Category)
**Master's/Integrated Master's-PhD Program/Integrated Bachelor's-Master's
Program/PhD Course**
End-Semester Examination-Autumn 2022

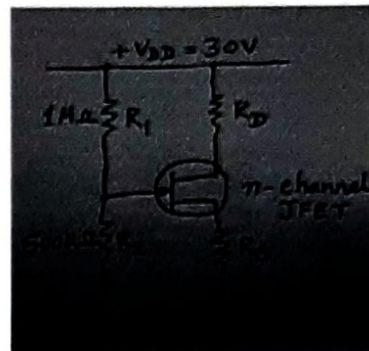
Subject: Electronics
Full Marks: 50

Subject Code(s): PHS 4104/AIS 4113/PHD 117
Time Allotted: 3 h

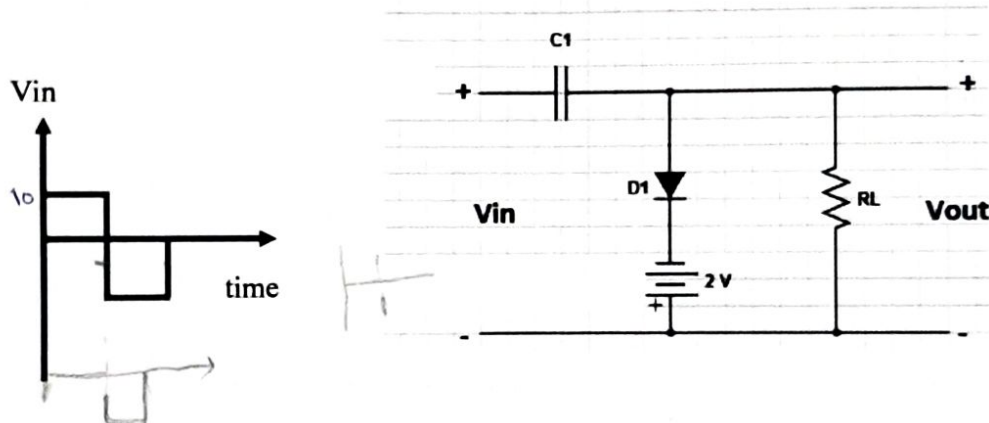
Answer all the following 8 questions

1. GaAs is a direct band gap semiconductor with $E_g = 1.42$ eV at 300 K. Take $N_c = N_v = 5 \times 10^{18} \text{ cm}^{-3}$ and independent of temperature. Calculate the intrinsic carrier concentration at room temperature. Explain numerically how the carrier concentration can be doubled without adding dopants. (2.5 + 2.5 marks)
2. Show schematically the junction formation between n-type semiconductor and metal when $\phi_m < \phi_s$ where ϕ_m and ϕ_s are work functions for metal and n-type semiconductor, χ is electron affinity and E_g is band gap of n-type semiconductor. (2.5 + 2.5 marks)
 - a. Determine whether the junction Schottky or Ohmic
 - b. Derive the expression for barrier height in equilibrium condition.
3. A p-i-n diode is made of a p-type, intrinsic, and n-type semiconductor (of the same material) joined together. Draw a qualitative energy band diagram if the p-i-n diode in equilibrium, forward bias, and reverse bias. Explain under which bias condition p-i-n diode has maximum current. (2.5 + 2.5 marks)
4. What is a tunnel diode, draw band bending diagram for a tunnel diode at various applied biases and explain the principle and mechanism of negative differential resistance in the tunnel diode via plotting I-V characteristics? (2.5 + 2.5 marks)
5. A solar cell at room temperature is under an illumination of 500 W.m^{-2} and has a short circuit current, I_{sc} , of 150 mA and an open circuit voltage, V_{oc} , of 0.53 V. What are the short circuit current and the open circuit voltage when the light intensity is doubled? What are the values when the intensity is halved? (2.5 + 2.5 marks)

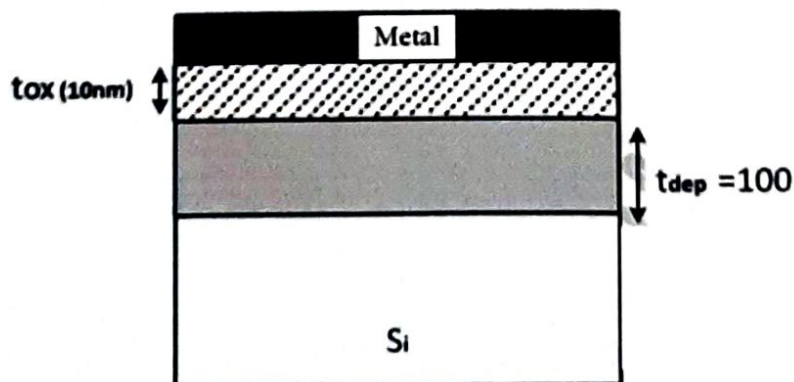
6. An n-channel JFET biased by the potential divider method, it is desired to set the operating point at $I_D = 2.5 \text{ mA}$ and $V_{DS} = 8 \text{ V}$. If $V_{DD} = 30 \text{ V}$, $R_1 = 1 \text{ M}\Omega$ and $R_2 = 500 \text{ k}\Omega$, find the value of R_S . The parameters of JFET are $I_{DSS} = 10 \text{ mA}$ and $V_{GS(off)} = -5 \text{ V}$. Nothing specified for R_D . (10 marks)



7. Sketch the output waveform in the same scale as the input for the circuit shown in the following figure (consider ideal diode). It is given that discharging time constant ($= CR_L$) is much greater than the time period of input wave. V_{in} is 10V peak-to-peak square wave input as shown below. (8 marks)



8. In a Metal Oxide Semiconductor capacitor (MOS) with an oxide layer thickness (t_{ox}) of 10 nm, the maximum depletion layer thickness ($t_{dep(max)}$) is 100 nm. The permittivity of the semiconductor and the oxide layer are ϵ_s and ϵ_{ox} respectively. Assuming $\epsilon_s/\epsilon_{ox} = 3$, find the ratio of the maximum capacitance and the minimum capacitance of this MOS capacitor. Please follow the figure for reference. (7 marks)



$$C_2 = \frac{A\epsilon_s}{d_2}$$

$$C_1 = \frac{A\epsilon_{ox}}{d_1}$$

$$C_1 = \frac{C_1 R_2}{C_1 + R_2}$$

$$\frac{C_{max}}{C_{min}} = \frac{C_1}{\frac{C_1 R_2}{C_1 + R_2}}$$

$$\frac{C_1 + R_2}{R_2}$$



Indian Association for the Cultivation of Science
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Master's/Integrated Master's - PhD/Integrated Bachelor's-Master's Program

End Semester Examination - Autumn 2022

Subject: Quantum Mechanics - I
Full marks: 50

Subject Code(s): PHS 4102/AIS 4111
Time allotted: 3 hrs

Attempt any five questions

1. (a) Consider an oscillator with electric charge e , placed in an external uniform electric field $\vec{E}$. Write the total Hamiltonian of the system. Calculate the first and the second order corrections to the n -th energy eigenvalue due to the external electric field. (5 marks)

(b) Show that the first order correction to the n -th energy eigenstate is,

$$|\psi_n^{(1)}\rangle = \frac{eE}{\hbar\omega} \sqrt{\frac{\hbar}{2m\omega}} [\sqrt{n}|n-1\rangle - \sqrt{n+1}|n+1\rangle]$$

where, the symbols have their usual meaning.

(5 marks)

2. (a) Show that the correction of the Hydrogen atom Hamiltonian due to the spin-orbit interaction is given by,

$$\hat{H}' = \frac{e^2}{2m_e^2 c^2} \frac{\vec{S} \cdot \vec{L}}{r^3}$$

where, symbols have their usual meaning.

(5 marks)

- (b) Show that $\hat{H}'$ is diagonal in the representation, which involve common eigenstates of the following operators: $\hat{L}^2$, $\hat{S}^2$, $\hat{J}^2$ and $\hat{J}_z$. (5 marks)

3. (a) Show that the first order relativistic correction to the Hydrogen atom is given by,

$$\hat{H}' = -\frac{\hat{p}^4}{8m_e^3 c^2}$$

where, symbols have their usual meaning.

(5 marks)

- (b) A strong magnetic field is applied to the Hydrogen atom along the z-direction. Show that,

$$\hat{H} = \hat{H}_0 + \frac{eB}{2m_e c} (\hat{L}_z + 2\hat{S}_z)$$

where, $\hat{H}_0$ is the unperturbed Hamiltonian of the Hydrogen atom. Find out the first order correction to the energy eigenvalues. (5 marks)

4. (a) Consider a harmonic oscillator Hamiltonian $\hat{H} = (\hat{p}^2/2m) + (1/2)m\omega_0^2\hat{x}^2$. Consider a trial wavefunction, $\psi(x) = Ae^{-\alpha|x|}$, where A is the normalization constant and α is the variational parameter. Find the variational energy $E(\alpha)$ of the oscillator, as a function of the parameter α . (5 marks)
- (b) Minimize $E(\alpha)$ with respect to α and find the best estimate to the ground state energy of the particle. What is the value of the optimal α ? (5 marks)
5. (a) A system in initial state $|i\rangle$ is perturbed by a harmonic Hamiltonian, $\hat{H}' = \hat{a}e^{i\hbar A} + \hat{a}^\dagger e^{-i\hbar A}$, over a time interval t . Show that the transition probability amplitude to a final state $|f\rangle$ is given by,

$$P_{if}(t) = \frac{4}{\hbar^2} \left[|\langle f|\hat{a}|i\rangle|^2 \frac{\sin^2\left(\frac{(\omega_{fi}+\omega)t}{2}\right)}{(\omega_{fi}+\omega)^2} + |\langle f|\hat{a}^\dagger|i\rangle|^2 \frac{\sin^2\left(\frac{(\omega_{fi}-\omega)t}{2}\right)}{(\omega_{fi}-\omega)^2} \right]$$

where, $\omega_{fi} = \omega_f - \omega_i$. From the plot of the above expression, explain the energy-time uncertainty relation. (5 marks)

- (b) Taking $t \rightarrow \infty$ in the above expression, show that the transition amplitude becomes,

$$P_{if} = \frac{2\pi}{\hbar} [|\langle f|\hat{a}|i\rangle|^2 \delta(E_f - E_i + \hbar\omega) + |\langle f|\hat{a}^\dagger|i\rangle|^2 \delta(E_f - E_i - \hbar\omega)]$$

Interpret the above result. (5 marks)

6. (a) Show that for scattering of a particle of mass m in a potential $V(r)$, the scattering amplitude in the first Born approximation is given by,

$$f_B(\theta) = -\frac{1}{4\pi} \left(\frac{2m}{\hbar^2} \right) \int e^{i(\vec{k}_i - \vec{k}_f) \cdot \vec{r}} V(r) d^3\vec{r}$$

where, symbols have their usual meaning. (5 marks)

- (b) For $V(r) = V_0 e^{-\mu r}/r$, show that the differential scattering cross-section for α particle scattering is,

$$|f_B(\theta)|^2 = \frac{(2Ze^2)^2}{16E^2 \sin^4(\theta/2)}$$

where, Z is the atomic number of the nucleus at the scattering centre. (5 marks)



Indian Association for the Cultivation of Science
(Deemed to be University under the *de novo* category)
BS-MS and Master's/Integrated Master's – PhD program
End-Semester Examination
Subject: Mathematical Methods-I
Subject Code(s): PHS4103, MAS4103P, AIS4110

Full marks: 50

Time allotted: 3 hrs

1. Use Kronecker delta and Levi-Civita symbols to prove the following identity for the vectors **A**, **B**, **C**.

✓ (a) $\text{div}(\mathbf{A} \times \mathbf{B}) = \mathbf{B} \cdot (\text{curl} \mathbf{A}) - \mathbf{A} \cdot (\text{curl} \mathbf{B})$

(b) Show

$$\frac{1}{2} \oint_C \vec{r} \times d\vec{r} = \vec{A}$$

where $\vec{A}$ is the vector area enclosed by the plane curve

✓ (c) Consider the vector field $\mathbf{u}(\mathbf{r}) = \mathbf{a}e^{i\mathbf{k} \cdot \mathbf{r}}$ where **a** and **k** are constant vectors. Let **V** and **S** denote the volume and surface of a sphere of radius **R** centered at origin. Show that:

$$\oint_S \mathbf{u} \cdot d\mathbf{S} = 4\pi R^3 (i\mathbf{k} \cdot \mathbf{a}) \frac{(\sin(kR) - kR\cos(kR))}{(kR)^3}$$

2+2+4

2(a) Find eigenvalues and eigenvectors of the following matrix.

$$A = \begin{pmatrix} 0 & 0 & -i \\ 0 & 2 & 0 \\ i & 0 & 0 \end{pmatrix}$$

✓ (b) The eigenvalues of a 3×3 matrix **M** are given to be 0, $i\alpha$ and $-i\alpha$, where α is a positive constant.

Using the Cayley-Hamilton theorem show that

$$\exp(iM) = I + i \frac{\sinh \alpha}{\alpha} M + \frac{(1 - \cosh \alpha)}{\alpha^2} M^2$$

✓ (c) The electromagnetic field strength tensor $F_{\mu\nu}$ is expressed in terms of the 4-vector potential A_μ by $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$. Construct $\bar{F}^{\alpha\beta} = \epsilon^{\alpha\beta\mu\nu} F_{\mu\nu}$.

✓ (f) Show that $\bar{F}^{\alpha\beta}$ is a contravariant tensor of rank two.

✓ (ii) Show that $\bar{F}$ satisfies $\partial_\alpha \bar{F}^{\alpha\beta} = 0$.

(d) A linear operator **A** is defined by:

$$A|e_1\rangle = |e_2\rangle; \quad A|e_2\rangle = |e_3\rangle; \quad A|e_3\rangle = |e_1\rangle$$

Find the matrix representation of **A**. What are the eigenvalues and eigenvectors of **A**? Is **A** orthogonal? what is the matrix representation of **A** in the basis:

$$|e'_1\rangle = \frac{|e_1\rangle + |e_2\rangle}{\sqrt{2}}$$

$$|e'_2\rangle = \frac{|e_1\rangle - |e_2\rangle}{\sqrt{2}}$$

$$|e'_3\rangle = |e_3\rangle$$

What is the transformation matrix **S** that takes from the unprimed co-ordinate system to the primed one?

✓ (e) Using the Gram-Schmidt orthogonalization procedure, construct the lowest two Hermite polynomials:

$$u_n(x) = x^n; \quad n = 0, 1, 2, \dots, \quad -\infty < x < \infty, \quad w(x) = e^{-x^2}$$

For this set of polynomials the usual normalization is

$$\int_{-\infty}^{\infty} H_m(x) H_n(x) w(x) dx = \delta_{mn} 2^m m! \pi^{1/2}$$

4+3+3+5+3

3(a) Consider the Bessel function $J_\nu(x)$ of order ν (real) given by:

$$J_\nu(x) = \sum_{r=0}^{\infty} \frac{(-1)^r}{r!(\nu+r)!} \left(\frac{x}{2}\right)^{\nu+2r}$$

(i) Show that the series for $J_{1/2}$ converges

(ii) Show the behaviour of $J_{1/2}$ near $x \simeq 0$

(iii) Derive the recurrence relations

$$\frac{d}{dx}(x^\nu J_\nu(x)) = x^\nu J_{\nu-1}(x)$$

$$\frac{d}{dx}(x^{-\nu} J_\nu(x)) = -x^{-\nu} J_{\nu+1}(x)$$

(iv) Show $J'_0 = J_1$

(b) (i) Show that the Wronskian $W(z)$ for a second order differential equation

$$y'' + p(x)y' + q(x)y = 0$$

is given by

$$W(z) = W(z_0) \exp\left(-\int_{z_0}^z p(z') dz'\right)$$

(ii) Find the Wronskian (up to an arbitrary constant) for the two linearly independent solutions of the Bessel equation.

$$y'' + \frac{1}{z}y' + \left(1 - \frac{\nu^2}{z^2}\right)y = 0$$

Does the Wronskian depend upon ν ?

(c) Verify the linear independence of the first three Hermite polynomials.

$$H_0 = 1, H_1 = 2x, H_2 = 4x^2 - 2$$

(d) Use the generating function

$$\phi(x, h) = \exp(2xh - h^2) = \sum_{n=0}^{\infty} H_n(x) \frac{h^n}{n!}$$

to prove

$$(i) H'_n(x) = 2nH_{n-1}(x)$$

$$(ii) H_{n+1}(x) = 2xH_n(x) - 2nH_{n-1}(x)$$

6+5+2+4

4(a) Expand $f(z) = \frac{1}{(z+1)(z+3)}$ in a Laurant series valid for $1 < |z| < 3$.

(b) Evaluate by the residue theorem: $\int_0^\infty \frac{dx}{1+x^4}$

2+5



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Master's/Integrated Master's-PhD Program/Integrated Bachelor's-Master's
Program

End-Semester Examination - Autumn 2022

Subject: Classical Mechanics
Full Marks: 50

Subject Code(s): PHS 4101/AIS 4110
Time Allotted: 3hr

Instructions

- (a) Attempt all questions.
- (b) For most of the questions the algebra will be self-explanatory; when some interpretation/description is needed, you can keep it brief but clear. There is no need to provide extensive description.
- (c) The marks for each question is given against the question. If you find that you cannot answer part (a), say, of a question but can answer part (b) *assuming the result in part (a)*, you may do so and you will get credit for part (b).

1. Consider the Lagrangian $L = -(m/2)q\ddot{q}$. Vary the action constructed from this Lagrangian and obtain the equation of motion for q . Hence, or, otherwise, find out the connection of this Lagrangian to that of a free particle Lagrangian. Compare the boundary conditions arising out of the variation of these two Lagrangians. (5 marks)
2. Argue that symmetries of space and time, along with Galilean invariance, is sufficient to derive the Lagrangian of a free particle. (5 marks)
3. Write down the Lagrangian for a double pendulum system and derive the equations of motion. Find out the Lagrangian in the small oscillation limit and determine the normal mode frequencies. (5 marks)
4. Derive the second law of Kepler from the Lagrangian of a particle moving in a central force field. Show that this law holds irrespective of the nature of the central force field. (5 marks)
5. A Hamiltonian with one degree of freedom q has the form,

$$H = \frac{p^2}{2\alpha} - bqpe^{-\alpha t} + \frac{ba}{2}q^2e^{-\alpha t}(\alpha + be^{-\alpha t}) + \frac{kq^2}{2},$$

where, α , a , b and k are constants. Find out the Lagrangian corresponding to the above Hamiltonian. Also determine another 'equivalent' Lagrangian, which is not an explicit function of time. Find out the Hamiltonian corresponding to this 'equivalent' Lagrangian and show how it differs from the origin Hamiltonian described above. (10 marks)

6. Write down the Hamiltonian of a harmonic oscillator and find out the canonical transformation to new coordinate and momenta, such that the system becomes cyclic. (5 marks)
7. Show that the Poisson bracket between three phase space variables, satisfy the following Jacobi identity, $[u[v, w]] + [w, [u, v]] + [v, [w, u]] = 0$. (5 marks)
8. Show that the function,

$$S = \frac{m\omega}{2}(q^2 + \alpha^2) \cot \omega t - m\omega q\alpha \operatorname{cosec} \omega t$$

is a solution of the Hamilton-Jacobi equation for a Harmonic oscillator. Further demonstrate that this function generates a correct solution to the motion of the harmonic oscillator in time. (10 marks)